

从标量碰撞行回到完整三维线性化： 受迫传递与 lift-up 边界

我从 Navier--Stokes 重新推导完整 Fourier--Leray row，固定 pressure factor two、Orr--Sommerfeld--Squire 符号、velocity recovery 和特殊零行。R0.72X 的 scalar invariant row 给出三种不同的 forcing scale；exact lift-up 解同时排除只依赖 ε_j 的 full-row strict contraction。完整 low-gap 向量行仍未闭合。

状态 · R0.72Y row ledger 完成

exact row identities and scoped scalar forcing

版本 v0.72Y · 2026-08-28

complete row algebra: CLOSED

standard H^{-1} spacetime: alpha

semiclassical H^{-1} spacetime: α^2

uniform full-row contraction: FALSE

strong full-row A2: OPEN

nonlinear / Clay: OPEN

00 / DIRECT DECISION

完整 row 恒等式和标量受迫尺度已闭合；完整强耦合行仍开放

CLOSED · EXACT ROW LEDGER

exactThreeDimensionalLinearization、
exactPressurePoissonFactorTwo、
exactOSSquireTriangularization、
exactVelocityReconstruction 与 fullRowEnergyIdentity 均为 CLOSED。

CLOSED · SCALAR FORCING

strongRowL2ForcingDuhamelAlpha2、
strongRowStandardHMinusOneTransferAlpha、
strongRowSemiclassicalHMinusOneTransferAlpha2 与
strongForcedDirectSumNoCountLoss 均为 CLOSED。

FALSE · UNSAFE EXTENSIONS

scalarA2EqualsCompleteRow、
epsilonOnlyFullRowClosure、
allPhysicalRowsUniformStrictContraction、
standardHMinusOneTransferAlpha2 与
HMinusOneEndpointAlphaGain 均为 FALSE。

OPEN · COMPLETE SYSTEM

strongFullRowA2Estimate、
scaleSharpOSPressureAbsorption、
orientationUniformSquireTransfer、
completeLinearizedShearSubsystem、
nonlinearNavierStokes 与 Clay 均为 OPEN。

压力右端的 factor two 来自两个不同散度项

$$\partial_t u + V \partial_{x_3} u + u_2 V_y e_3 + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0,$$

$$\Delta_K p = -2i K_z V_y u_2.$$

一份 $i K_z V_y u_2$ 来自 background transport，另一份来自 shear-gradient term；符号与 $+\nabla p$ 在左侧的约定一起冻结。

完整 row label 同时保留水平方向与 residue class

$$j = (K_x, K_z, [r]_R), \quad A_\beta = \partial_x + i\beta, \quad \mathcal{L} = -A_\beta^2 + \mu, \quad \mu = \xi^2 + \gamma^2,$$

$$\mathbb{P}_j = I + \nabla_j \mathcal{L}^{-1} \operatorname{div}_j, \quad u_d = -\mathcal{L} u - \mathbb{P}_j (icW u + \Lambda W_x u_2 e_3).$$

orthogonal direct sum 是结构恒等式；它不会自动生成 row-uniform bound。

$\mu>0$ 时得到精确三角系统

$$q = \mathcal{L} u_2, \quad \eta = i\gamma u_1 - i\xi u_3,$$

$$q_d = (-\mathcal{L} - icW)q - icW_{xx} \mathcal{L}^{-1} q,$$

$$\eta_d = (-\mathcal{L} - icW)\eta + i\xi \Lambda W_x \mathcal{L}^{-1} q.$$

pressure feedback 与 Squire transfer 的尺度不同； $\mu=0$ 不能套用这组逆公式。

速度恢复与 kinetic energy identity 均保留方向权重

$$u_1 = \frac{i}{\mu}(\xi A_\beta u_2 - \gamma \eta), \quad u_3 = \frac{i}{\mu}(\gamma A_\beta u_2 + \xi \eta),$$

$$\|u\|_2^2 = \|u_2\|_2^2 + \mu^{-1}(\|A_\beta u_2\|_2^2 + \|\eta\|_2^2).$$

Ro.72X scalar operator 是完整 row 的严格不变子空间

$$u = g(\gamma, 0, -\xi)/\sqrt{\mu}, \quad g_d = (A_\beta^2 - \mu)g - icWg.$$

scalarA2InvariantEmbedding=CLOSED; scalarA2EqualsCompleteRow=FALSE。一般数据仍会激活 pressure feedback 与 Squire transfer。

high-gap class 有显式衰减，low-gap class 没有被外推

$$\frac{1}{2} \frac{d}{dd} \|u\|^2 + \|A_\beta u\|^2 + \mu \|u\|^2 = -\Lambda \operatorname{Re} \langle W_x u_2, u_3 \rangle.$$

若 $g_j > |\Lambda| M_K/2$ ，则 norm exponent 是 $g_j - |\Lambda| M_K/2$ 。这是 damping-dominated full rows，不是完整 low-gap theorem。

zero coupling 仍可产生严格 transient growth

$$u_2(d) = e^{-\xi^2 d} v_0, \quad u_3(d) = -\Lambda d e^{-\xi^2 d} W_x(d, x) v_0,$$

$$\frac{\|u(d)\|^2}{\|u(0)\|^2} = e^{-2\xi^2 d} \left[1 + \frac{\Lambda^2 d^2}{8} (e^{-2d} + e^{-8d}) \right].$$

$\xi=0$ 时任意非零 Λ 都增长； $\xi>0$ 给 spatially mean-zero 扰动，取足够大的 $|\Lambda|$ 仍可增长。因此 `epsilonOnlyFullRowClosure=FALSE`, `allPhysicalRowsUniformStrictContraction=FALSE`。

08 / EXACT CAUSAL KERNEL

几何 block envelope 直接给出 scalar L2 forcing 的 alpha-squared scale

$$\int_0^\infty \left(e^{-\mu r} q^{\lfloor r/h \rfloor} \right)^p dr = \frac{1 - e^{-p\mu h}}{p\mu(1 - q^p e^{-p\mu h})}, \quad h = 2T\alpha^2.$$

$\mu \rightarrow 0$ 的极限是 $h/(1 - q^p)$ 。Young convolution 给 $L_d^2 L_x^2 \rightarrow L_d^2 L_x^2$ 的 $O(\alpha^2)$ bound。

09 / NEGATIVE SOBOLEV FORCING

standard 与 semiclassical negative norm 必须分开

$$\|G\|_{L_d^2 L_x^2} \leq C_q \alpha \|F\|_{L_d^2 H_\beta^{-1}},$$

$$\|G\|_{L_d^2 L_x^2} \leq C_q \alpha^2 \|F\|_{L_d^2 \mathcal{H}_{\alpha,\beta}^{-1}}.$$

两条结论来自 backward-adjoint energy 与 transposition；`standardHMinusOneTransferAlpha2=FALSE`。

10 / ENDPOINT BOUNDARY

standard endpoint 只有 scale one；semiclassical endpoint 有 alpha

$$\max\{\|G\|_{L_d^\infty L_x^2}, \|A_\beta G\|_{L_d^2 L_x^2}\} \leq C'_q \|F\|_{L_d^2 H_\beta^{-1}}.$$

semiclassical 输入的右端另有一个 α 。terminal high-frequency pulse 给 order-one endpoint response，所以 `HMinusOneEndpointAlphaGain=FALSE`。

collision-chart witness 分别饱和 α 与 α -squared

compactly supported mean-zero witness 的 scaling ratios 分别是 standard H^{-1} 的 α 和 semiclassical negative norm 的 α^2 ；证明使用解析缩放与极限，不由有限证书代替。

有限历史 energy ledger 闭合，但没有共同 strong scale

weak/zero scalar rows 有 $O_K(1)$ 的 finite-history bound；有 covariant gap 时还有 time-global estimate。gapless constant row 在持续 forcing 下线性增长。mean-zero 也不在 WG 下自动保持。

已有 forced 与 vector 先例，但现有方法没有覆盖本节组合

Wei--Zhang 已在 monotone Couette 邻域给出 nonautonomous forced 和三维向量闭合；该坐标法在 $V_y = 0$ 的 collision 处失效。Coble--He 的 fixed-shape homogeneous theorem 不含 forcing 或完整向量行。我只报告有界原始文献核验，不把“未发现”写成首创性证明。

有限代数、函数分析和附图各有明确职责

双路证书核对 heat identity、pressure factor、Leray signs、OS--Squire coefficients、velocity recovery、lift-up residual、kernel algebra 与 Fourier weights。Galerkin limit、duality、endpoint trace、sharpness limit 和 evolution-family existence 由解析报告承担。附图只可视化精确公式与已证明 power guides。

正式附图分开显示 row structure、lift-up 与 forcing powers

R0.72Y | full Fourier row and forced-transfer boundaries

Exact algebra, exact lift-up counterexample, and normalized forcing-rate guide



A Exact complete-row structure

The scalar invariant row closes; orthogonality alone does not close the full row.

Complete Fourier-Leray row

$$\begin{aligned} u_d &= -L u - P_j(\\ &\quad i c W u + \Lambda W_x u_2 e_3) \\ \text{div}_j u &= 0 \\ L p_i &= 2 i c W_x u_2 \end{aligned}$$

pressure factor 2: CLOSED

exact

$\mu > 0$: Orr-Sommerfeld / Squire

$$\begin{aligned} q_d &= (-L - i c W) q \\ &\quad - i c W_{xx} L^{-1} q \\ \eta_d &= (-L - i c W) \eta \\ &\quad + i \xi \Lambda W_x L^{-1} q \end{aligned}$$

lift-up forcing

q

eta

STRONG FULL-ROW A2: OPEN

Scalar invariant polarization

$$\begin{aligned} u &= g(\gamma, 0, -\xi) / \sqrt{\mu} \\ g_d &= (A_{\beta}^2 - \mu - i c W) g \end{aligned}$$

scalar A2 embedding: CLOSED

$K_z=0$ exact lift-up

$$u_3 = -\Lambda d \exp(-\xi^2 d) \times W_x(d, x) u_2(0)$$

uniform strict contraction: FALSE

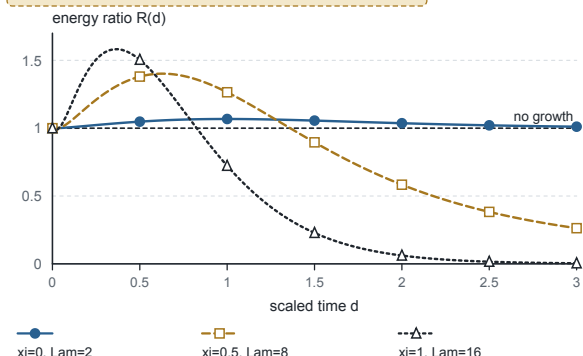
Velocity recovery is exact only for $\mu > 0$; the $\mu=0$ row is checked directly.

Direct-sum orthogonality removes row-count loss only after a uniform row bound; it does not create that bound.

B Exact zero-coupling lift-up

Energy ratio from the closed formula; line at 1 means no growth.

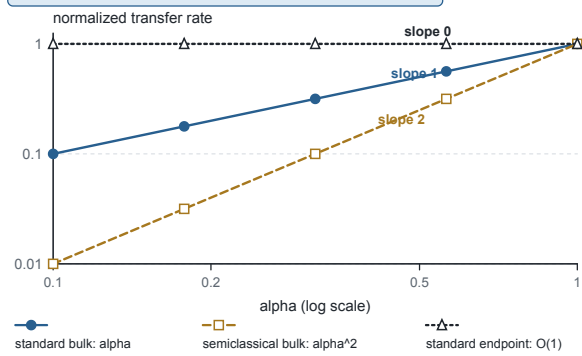
EXACT COUNTEREXAMPLE - NOT A STABILITY PROOF



C Scalar forced-transfer powers

Normalized operator-rate guide; constants are suppressed.

EXACT RATE GUIDE - ANALYTIC PROOF ELSEWHERE



R0.72Y | exact presentation sampling; proof and sharpness are audited separately

[下载 PDF](#) · [下载 PNG](#) · [打开 SVG](#)

16 / RESEARCH VALUE

本节把 scalar result 放回正确的完整线性边界

严格价值在于不再把 scalar A2 estimate 与 complete vector row 混同，并固定了 forcing topology 对 α power 的影响。直接 Clay 价值仍低：low-gap vector estimate、nonlinear convolution、vortex stretching bootstrap 与 continuation criterion 都未完成。

17 / NEXT GATE

Ro.72Z: 吸收 Orr--Sommerfeld feedback, 并支付 orientation-dependent Squire transfer

下一节直接研究 $q \rightarrow \eta$ 三角系统，按 $|\xi/\gamma|$ 、damping gap 与 $\mu = 0$ exceptional rows 分区；任何结论都必须保留 lift-up transient prefactor。

完整报告、边界矩阵、独立审计、证书与附图

[完整数学报告](#) · [文献边界审计](#) · [主张—证据矩阵](#) · [独立数学审计](#)

[确定性双路证书](#) · [正式附图包](#) · [同步研究笔记 PDF](#) · [累计回顾](#) · [累计回顾 PDF](#)